

Maths

Unit 1.1 – Place Value

WALT: Compose, decompose and regroup numbers including decimals.

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**I can compose
and decompose
decimal
numbers.**



**I can use
regrouping to
solve missing
number
questions
involving
decimals.**



**I can use
regrouping to
decompose a
number in
multiple ways.**

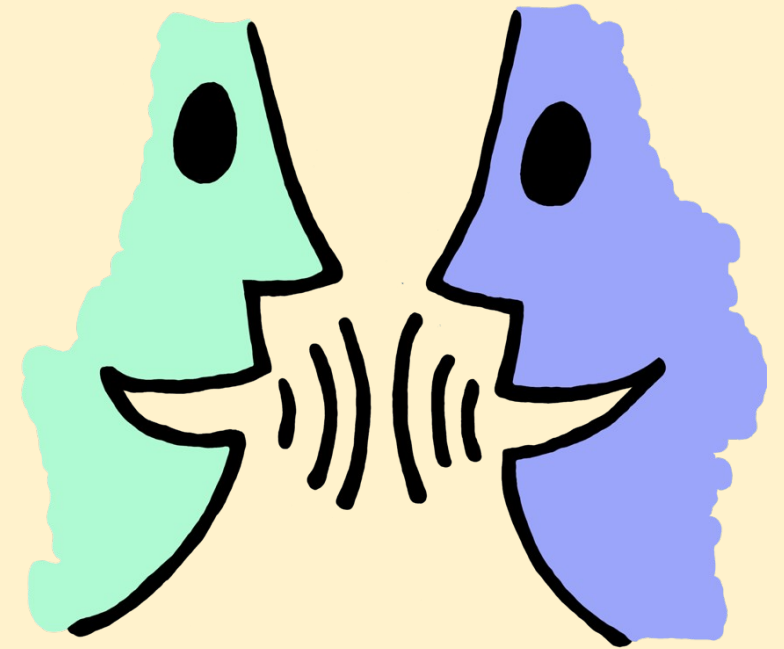
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Retrieval Practice -

Ms. Colette says “5,807 divided by 1,000 is 5.87”

Is she correct?

What mistake has she made?



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Starter - What do the following words mean?

Compose

Put together.

Decompose

Break down into parts.

Regroup

Change the way a number is written.

WALT: Compose, decompose and regroup numbers including decimals.

We can use our knowledge of place value to **decompose** decimal numbers.

34.832

30 + 4 + 0. + 0.0 + 0.00

8 3 2

How have we decomposed this number?

It has been decomposed into its place value.

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We can use our knowledge of place value to **decompose** decimal numbers.

34.832

30 + 4 + 0. + 0.0 + 0.00

8 3 2

Is this the only way to decompose the number?

No, but we call this regrouping which we will look at later.

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We can use our knowledge of place value to **compose** decimal numbers.

$$400 + 30 + 2 + 0.9 + 0.01$$

=

$$432.91$$

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Check your understanding.

Compose a number using these parts.

$$500 + 8 + 0.05 + 0.009 = 508.059$$

Decompose this number into its parts.

650.403

$$600 + 50 + 0.4 + 0.003$$

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Regrouping is when we exchange different amounts between different place values.

42.329

$40 + 2 + 0.3 + 0.02 + 0.009$

$30 + 12 + 0.3 + 0.02 + 0.009$

$42 + 0.1 + 0.22 + 0.009$

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Compose these parts to make a number.

a) $90 + 7 + 0.1 + 0.05$

b) $500 + 3 + 0.04 + 0.003$

c) $3,000 + 20 + 0.4 + 0.001$

Decompose these numbers into their parts.

d) 425.25

e) 92.086

g) 210.064



Use your knowledge of regrouping to find the missing numbers.

e) $773.23 = 700 + 70 + \underline{\hspace{2cm}} + 0.2 + 0.03$

f) $0.582 = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + 0.002$

g) $2.405 = 2 + 0.4 + \underline{\hspace{2cm}}$

h) $0.934 = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + 0.004$

i) $57.12 = 40 + \underline{\hspace{2cm}} + 7 + 0.1 + \underline{\hspace{2cm}}$

j) $219.901 = 100 + \underline{\hspace{2cm}} + 9 + \underline{\hspace{2cm}} + 0.001$



k) Conject: How many different ways can you regroup this number?

782.493

Now write as many different ways as you can.

Compare your answers with someone else.
Was your conjecture correct?

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Compose these parts to make a number.

a) $90 + 7 + 0.1 + 0.05$

b) $500 + 3 + 0.04 + 0.003$

c) $3,000 + 20 + 0.4 + 0.001$

a) 97.105

b) 503.043

c) 3,020.401

Decompose these numbers into their parts.

d) 425.25

e) 92.086

g) 210.064

d)

$400 + 20 + 5 + 0.2 + 0.05$

e) $90 + 2 + 0.08 + 0.006$

g)

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Use your knowledge of regrouping to find the missing numbers.

e) $773.23 = 700 + 70 + 3 + 0.2 + 0.03$

f) $0.582 = 0.5 + 0.08 + 0.002$

g) $2.405 = 2 + 0.4 + 0.005$

h) $0.934 = 0.9 + 0.03 + 0.004$

i) $57.12 = 40 + 10 + 7 + 0.1 + 0.02$

j) $219.901 = 100 + 110 + 9 + 0.9 + 0.001$

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i) Conject: How many different ways can you regroup this number?

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Plenary - Two stars and a wish.

Tell your partner two things that went well and one thing that you want to do in the future regarding today's lesson.

